# Mohammad Shahnawaz

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## Skills

**Languages:** Python (primary), JavaScript, SQL
**AI/ML:** LangChain, LangGraph, Vector Databases(ChromaDB), LLMs, RAG, Embeddings, Prompt Engineering
**Web:** FastAPI, Flask, REST APIs, Pydantic, Asynchronous Programming, React, Supabase
**Systems & Infra:** PostgreSQL, Docker, CI/CD, Linux, Railway, Vercel, Hugging Face, Redis
**Developer Tools:** Git, GitHub, Pytest, VS Code, Cursor

## Experience

**AI Engineering Intern | Flyrank.ai** July 2026 – Oct 2026

- Built an AI assisted chatbot with the response grounded to 50+ documents (FAQs) using RAG based system.
- Designed a schema-first RAG pipeline routing typed field queries to direct lookups and unstructured queries to ChromaDB via **lfm-2.5-embedding** embeddings, eliminating semantic search overhead for exact-match questions.
- Evaluated retrieval quality and improved **recall@5** from 40% to 75% using hybrid retrieval (semantic and BM25) with a cross-encoder Reranker over 70 query **golden dataset**.

**Software Developer (Contract) | Raza Studioz** Dec 2025 – Feb 2026

- Built a full-stack web app with React, serverless REST API, and Supabase (PostgreSQL) serving 100+ active users.
- Engineered an email dispatch system with Resend for real-time admin notifications (sub-2s delivery), while collaborating with the founder to prioritize features under shifting deadlines.

**Open Source Contributor | MetaBrainz Foundation (ListenBrainz)** Feb 2025 – May 2025

- Persisted the music player's volume state via localStorage in the React frontend, retaining user preference across sessions for better user experience.
- Contributed 4+ pull requests reviewed and merged by core maintainers while iterating across multiple review cycles.

## Projects

**MedAI** | *FastAPI, LangChain, Gemini, PostgreSQL* [Live Demo], [GitHub]

- Architected an agentic chatbot for booking appointments, checking availability related queries using LangGraph agent orchestration; added **8 tools** accessing database (PostgreSQL) and email notification with Pydantic schemas
- Constrained agents with routing limits, tool restrictions, input guardrails (length of query) and maximum of 3 retry logic for failed tool call to prevent agent stuck in loop.

**ResumeXAnalyzer** | *FastAPI, LangChain, Railway, React* [Live Demo], [GitHub]

- Developed ResumeXAnalyzer, a multi-agent AI platform that helps students and job seekers identify resume gaps and improve job-fit alignment and analyzed 50+ resumes with sub-15s average processing time.
- Designed multiple versions of system prompts and evaluated 50 query for better response quality using LLM as judge.

**Robustness Analysis of vision models** | *Python, YOLO model, NumPy, Ultralytics* [GitHub]

- Evaluated YOLOv8–YOLO26 object detection model under Gaussian, Salt-Pepper and Poisson noise.
- Evaluated models across Precision, Recall, mAP, F1-Score, Latency, and a custom Robustness Retention Index to quantify performance degradation.
- Identified YOLOv9s as the most noise-robust model and YOLOv10n as the best overall performer, guiding model selection for real-time, real-world deployment

## Education

**Guru Ghasidas Vishwavidyalaya (Bilaspur)** Nov 2022 – May 2026
*B.Tech in Electronics and Communication Engineering* *CGPA: 8.10/10*

## Certifications and Achievements

- Completed CS50's AI with Python from Harvard University. [Certificate]
- Completed CS50x – Introduction to Computer Science from Harvard University. [Certificate]
- Gold Medalist (Top 1% among 33,131) in Introduction to Internet of Things (IIT Kharagpur). [Certificate]